

declarative message routing to other Azure services, and provides extensive monitoring for device connectivity and device identity management events.

- **Azure API Management:** This is a solution for publishing APIs to external and internal customers. It quickly creates consistent and modern API gateways for existing back-end services hosted anywhere; it secures and protects them from abuse and overuse and gets insights into usage and health. Azure API Management provides the core competencies to ensure a successful API program through developer engagement, business insights, analytics, security and protection.
- **Azure IoT Edge:** This is an open source solution from Microsoft Azure for edge computing. It offers device management, compute, analytics, operating with offline connectivity and real-time decision making at the edge of an IoT solution. And it helps to reduce the bandwidth costs incurred for data communication to the cloud from the edge. It is a service that delivers cloud capabilities to the edge. Azure IoT Edge provides easy orchestration between code and services, so they flow securely between the cloud and the edge to distribute intelligence across IoT devices. It easily integrates Microsoft Azure and third-party services or augments existing services to create a custom IoT application with business logic. It helps to put intelligence into devices which can act locally, based on the data they generate, while also taking advantage of the cloud to configure, deploy and manage these devices securely and at scale.
- **AWS IoT:** This is a managed cloud platform that lets connected devices easily and securely interact with cloud applications and other devices. It can support billions of devices and trillions of messages. It can process and route those messages to AWS endpoints and to other devices reliably and securely. With AWS, IoT applications can keep track of and communicate with all devices, all the time, even when they aren't connected.
- **AWS API Gateway:** This is a fully managed service that makes it easy for developers to create, publish, maintain, monitor and secure APIs at any scale. With a few clicks in the AWS Management Console, the user can create an API that acts as a 'front door' for applications to access data, business logic, or functionality from back-end services. It handles all the tasks involved in accepting and processing up to hundreds of thousands of concurrent API calls, including traffic management, authorisation and access control, monitoring, and API version management.
- **AWS Greengrass:** This is a service from AWS for edge computing. It offers local compute, messaging, data caching, sync and machine learning capabilities for connected devices. It ensures quick responses to local events and actions along with local storage, and helps to minimise the cost of transmitting IoT data to the cloud.

Admin information dashboard

Both the leading cloud service providers facilitate administrative consoles to configure, operate, view and manage the IoT devices and data protocols. IoT solutions powered by cloud services can also use these consoles.

Other IoT tools and services

Some of the other IoT specific tools, technologies or services from Azure/AWS are AWS IoT SDK, AWS IoT Registry, AWS IoT Security, AWS Rule Engine, AWS Device Shadows, Azure IoT Device SDK, Azure IoT Protocol Gateway, Azure AD and Azure Device Provisioning.

Back-end technologies

An end-to-end IoT solution can use many other services from cloud service providers like Azure Data Factory, Azure Stream Analytics, Azure Storage, Azure Cosmos DB, AWS Kinesis, AWS Data Pipeline, AWS SQS, AWS DynamoDB and AWS S3.

Note: Do refer to the AWS and Azure sites for the latest updates on their respective service details.

Reference architecture of edge computing for IoT solutions with the open source technology stack

The following are the technologies/frameworks available.

- **Kura:** Kura is a Java/OSGi-based open source framework for IoT systems. It has support to access the underlying hardware like serial ports, GPS, watchdog, etc. It supports the management of IoT devices, including configurations and communication. It provides M2M/IoT integration platforms and gateway management. Kura also provides various APIs, interfaces and capabilities for communication, connectivity, network management, data management, messaging, remote management, etc.
- **Kapua:** This is an IoT platform to manage and integrate devices and their data. It provides an integration framework and other features like a device registry, device management, messaging services, data management and application enablement.
- **Spring Boot:** This is a spring framework for the development of REST APIs. It facilitates the creation of standalone, production-grade spring based applications. It works with minimum spring configurations.

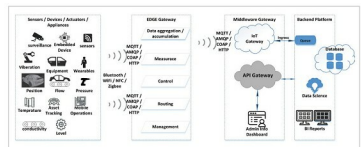


Figure 4: Reference architecture for open source